Chapter 12, Problem 78CP

Problem

A three-phase 440-V, 51-kW, 60-kVA inductive load operates at 60 Hz and is wye-connected. It is desired to correct the power factor to 0.95 lagging. What value of capacitor should be placed in parallel with each load impedance?

Step-by-step solution

Step 1 of 2

$$Cos\,\theta_1 = \frac{51}{60} = 0.85 \Rightarrow \theta_1 = 31.79°$$

$$\theta_1 = S_1\,Sin\,\theta_1 = (60)(0.5268) = 31.61\ KVAR$$

$$P_2 = P_1 \quad 51KW$$

$$Cos\,\theta_2 = 0.98 \Rightarrow \theta_2 = 18.19°$$

$$S_2 = \frac{P_2}{Cos\,\theta_2} = 53.68\ KVA$$

$$\theta_2 = S_2\,Sin\,\theta_2 = 16.759\ KVAR$$

$$\theta_c = \theta_1 - \theta_2 = 3.61 - 16.759 = 14.851\ KVAR$$

Step 2 of 2

For each wad,

$$\theta_C = \frac{\theta_c}{3} = 4.95\ KVAR$$

$$C = \frac{\theta_c}{\omega V^2} = \frac{4950}{(2\pi)(60)(440)^2}$$

$$= 67.82\ \mu F\ .$$